Q.30 Explain the series and parallel combination of resistance with circuit diagram. (CO4)

Q.31 Write the care and maintenance of lead acid batteries. (CO3)

Q.32 Difference between electrical and magnetic circuits. (CO6)

Q.33 Explain the Fleming's right hand rule. (CO5)

Q.34 What is the power factor in AC circuits? State disadvantage of low power factors. (CO7)

Q.35 Discuss about parallel resonance in AC circuit. (Co7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain Thevenin Theorem and its use in electric circuit. (CO2)

Q.37 State and explain Faraday's law of electro-magnetic induction. (CO4)

Q.38 Explain the construction, principle and working of lead-acid battery. (CO5)

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Roll No. ....

2nd Semester/ Comp, ECE, IT, I & Control, Med, Eltx, Eltx & Instr., Power Eltx, EEE
Subject : Basic Electrical Engineering

Time : 3 Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Internal resistance of ideal voltage source is (CO3)

a) Zero b) Infinite

c) Very Low d) Very High

Q.2 The frequency of DC is (CO6)

a) Zero b) 50

c) 100 d) Infinite

Q.3 The number of cycles completed in one second is called (CO6)

a) Time Period b) Voltage

c) Frequency d) None of above

Q.4 The unit of current is (CO1)

a) Ampere b) Volt

c) Watt d) None of these

Q.5 Power factor for pure inductive load is (CO3)

a) Unity b) Zero

c) Lagging d) None of these

Q.6 The unit of inductance is (CO7)

a) Ohm b) Watt

c) Volt d) Henery

Q.7 Unit of flux density is (CO4)

a) Henery b) Weber / m^2

c) Joule d) None of the above

Q.8 The rate of doing work is called (CO1)

a) Power b) Current

c) Voltage d) None of these

Q.9 The energy meter measures the energy in (CO2)

a) Watts b) KWH

c) Megawatts d) Kilowatts

Q.10 The solar cell converts solar energy into ________ energy. (CO5)

a) Chemical b) Mechanical

c) Electrical d) None of above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 Insulators have __ resistance. (CO1)

Q.12 Define KCL? (CO2)

Q.13 What is mutual induction? (CO4)

Q.14 What is the unit of capacitance? (CO3)

Q.15 Define Susceptance. (CO1)

Q.16 What are passive components? (CO1)

Q.17 In a circuit power is measured by _____. (CO2)

Q.18 The device which converts DC into AC is called_____. (CO1)

Q.19 Unit of conductance is _____. (CO4)

Q.20 What is the full form of EMF? (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain maximum power transfer theorem? (CO2)

Q.22 Discuss concept of Ideal Current Source. (CO3)

Q.23 Give significance to the power factor. (CO7)

Q.24 Explain Kirchhoff voltage law. (CO1)

Q.25 What is a cell? Explain its types. (CO5)

Q.26 Difference between a.c. and d.c. (CO1)

Q.27 What is Resonance? Explain series resonance in detail. (CO3)

Q.28 Explain the working of solar cell. (CO5)

Q.29 Define flux and reluctance. (CO4)